

PHUBAI, Viet Nam

WMO: 488520

Lat: 16.402N

Lon: 107.703E

Elev: 15

StdP: 101.15

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 14.2	(c) 15.2	(d) 13.0	(e) 9.3	(f) 15.9	(g) 13.9	(h) 10.0	(i) 16.2	(j) 6.5	(k) 21.9	(l) 6.0	(m) 22.1	(n) 1.8	(o) 270	(p) 0.266

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 9.3	(c) 37.2	(d) 27.2	(e) 36.2	(f) 26.9	(g) 35.2	(h) 26.7	(i) 28.8	(j) 34.3	(k) 28.1	(l) 33.5	(m) 27.6	(n) 32.8	(o) 3.6	(p) 50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.5	23.4	31.0	26.9	22.7	30.4	26.2	21.7	29.7	93.8	34.9	90.7	33.5	88.3	33.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.3	(b) 5.3	(c) 4.6	(e) 12.8	(f) 39.1	(g) 2.2	(h) 1.8	(i) 11.2	(j) 40.3	(k) 9.8	(l) 41.4	(m) 8.6	(n) 42.4	(o) 7.0	(p) 43.6		
(a) 6.3	(b) 5.3	(c) 4.6	(e) 12.4	(f) 29.7	(g) 2.2	(h) 1.4	(i) 10.9	(j) 30.8	(k) 9.6	(l) 31.6	(m) 8.4	(n) 32.4	(o) 6.8	(p) 33.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.9	20.5	21.6	23.9	26.9	29.0	30.1	29.8	29.7	28.0	26.2	24.1	21.3
	DBStd	4.13	2.70	3.37	3.30	2.77	2.07	1.53	1.45	1.56	1.71	1.67	2.34	2.68
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	30	11	10	3	0	0	0	0	0	0	1	6	
	CDD10.0	5817	327	326	430	508	589	603	614	609	539	501	422	349
	CDD18.3	2804	79	102	174	258	331	353	356	351	289	243	172	96
	CDH23.3	27581	259	512	1187	2458	3803	4543	4456	4254	2901	1894	982	333
CDH26.7	11799	37	140	388	1013	1759	2318	2203	2052	1102	532	208	49	
Wind	WSAvg	1.7	1.7	1.7	1.6	1.5	1.6	1.7	1.6	1.6	1.6	1.8	1.9	1.9
Precipitation	PrecAvg	3108	127	57	61	67	122	84	83	172	435	802	653	422
	PrecMax	5441	348	144	178	204	323	232	231	614	1303	1488	2220	1151
	PrecMin	2078	32	4	11	7	29	16	20	31	57	325	163	89
	PrecStd	794	88	37	52	54	71	54	58	133	269	310	431	256
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.9	33.0	34.6	37.0	38.1	39.0	38.2	38.7	36.2	34.1	32.0	30.0
		MCWB	24.4	25.3	26.4	27.8	27.8	27.5	27.7	27.1	27.0	27.0	26.2	25.2
	2%	DB	27.4	30.4	32.7	35.1	36.2	37.0	36.6	36.7	34.7	32.0	30.2	28.0
		MCWB	23.1	25.0	26.2	27.4	27.4	27.3	26.7	26.7	26.7	26.4	25.6	24.1
	5%	DB	26.0	28.1	30.8	33.7	35.1	35.8	35.3	35.2	33.2	30.7	29.0	26.3
		MCWB	22.5	24.1	25.8	26.9	27.1	26.9	26.3	26.4	26.5	25.8	25.0	23.1
	10%	DB	24.6	26.2	28.8	32.0	33.8	34.7	34.3	34.0	31.9	29.5	27.7	25.1
		MCWB	22.0	23.1	24.9	26.4	26.9	26.7	26.3	26.3	26.4	25.4	24.5	22.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.9	26.3	27.4	28.8	29.2	29.4	29.6	29.5	28.1	27.9	26.7	25.7
		MCDB	28.4	31.0	32.6	35.1	34.6	35.2	35.2	35.1	33.2	31.5	30.1	28.9
	2%	WB	23.9	25.2	26.6	27.8	28.2	28.4	28.2	28.1	27.5	26.9	26.1	24.7
		MCDB	26.4	29.2	31.2	33.4	34.1	33.9	33.1	33.1	32.1	30.5	29.0	26.9
	5%	WB	23.1	24.5	25.9	27.2	27.6	27.7	27.5	27.4	27.0	26.3	25.6	23.9
		MCDB	25.1	27.5	29.8	32.1	33.2	33.1	32.5	32.3	31.3	29.3	28.1	25.7
	10%	WB	22.4	23.6	25.2	26.6	27.1	27.3	27.0	26.9	26.6	25.9	25.0	23.1
		MCDB	24.0	25.8	28.2	30.9	32.3	32.5	32.1	31.7	30.4	28.5	27.1	24.6
Mean Daily Temperature Range	5% DB	MDBR	5.5	6.4	6.9	7.9	8.8	9.3	9.1	9.0	7.3	6.3	5.4	4.7
		MCDBR	8.4	9.9	10.2	10.8	11.0	11.2	10.9	11.0	9.8	8.7	7.8	7.5
		MCWBR	4.3	4.7	4.7	4.5	3.9	3.8	3.7	3.5	3.5	3.6	3.5	3.7
	5% WB	MCDBR	7.4	9.3	9.3	9.9	10.0	10.1	10.0	9.9	8.6	7.5	6.9	6.7
		MCWBR	4.1	4.6	4.5	4.5	4.0	3.9	3.9	3.7	3.4	3.4	3.4	3.6
Clear-Sky Solar Irradiance	taub	0.521	0.563	0.685	0.622	0.516	0.471	0.455	0.466	0.527	0.578	0.544	0.562	
	taud	2.016	1.934	1.691	1.873	2.177	2.320	2.358	2.314	2.132	1.945	2.005	1.898	
	Ebn at Noon	766	756	679	726	796	824	839	838	789	733	740	713	
	Edh at Noon	167	190	249	208	151	130	125	132	158	185	168	184	
All-Sky Solar Radiation	RadAvg	2.96	3.96	4.34	5.11	5.57	5.48	5.15	5.03	4.51	3.71	3.08	2.39	
	RadStd	0.45	0.74	0.42	0.46	0.41	0.35	0.52	0.37	0.54	0.54	0.58	0.61	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.48	N/A	N/A	+0.94	+0.53	+0.50	N/A	N/A	+302
(47) Regional (3 neighbors)		+0.38	N/A	N/A	+0.67	+0.30	N/A	N/A	N/A	+228

Nomenclature: See separate page